

A Comparative Study of Quantum and Classical Deep Learning for Intelligent Agriculture

Roshan Kumar Singh¹, Amreen Khan²

¹Junior Research Fellow

Electrical and Electronic Engineering Department BITS Pilani, Pilani- 333031

roshan2rise@gmail.com

²Research Scholar

Centre for Research in Nanotechnology and Science, Indian Institute of Technology Bombay,

Mumbai- 400076

amreen.khan.mtech@gmail.com

Abstract

Agriculture, playing a vital role in human life, constitutes over 60 percent of the economy in many developing countries. Despite its significance in providing food and employment, agricultural productivity often faces challenges in terms of efficiency. One major obstacle is the damage caused to grains due to plant diseases. To address this issue, modern technologies have been employed to monitor various factors and minimize damage. These technologies generate vast amounts of data, necessitating advanced analytical approaches for effective decision-making. Deep learning has already demonstrated its potential in revolutionizing various industries, including agriculture. By harnessing this technique, we aim to analyze and interpret the copious amounts of data generated through technological interventions, thereby making informed decisions to enhance agricultural productivity. While classical deep learning methods have been applied extensively in agriculture, the emergence of quantum deep learning opens new possibilities for addressing complex problems in finance, healthcare, and transportation. Satiating the demand, in this paper, we present two techniques namely classical and quantum deep learning for agriculture by comparing their performance.

KEYWORDS: Agriculture, production, deep learning, quantum deep learning

I. Introduction

Agriculture is not only an essential human need but also a significant contributor to the economy. The increasing global population, including in India, poses numerous challenges for the agricultural sector [1]. Agriculture highly depends on weather and climate change, affecting the productivity of grains. Rainfall, soil nutrients, humidity, solar radiation, and temperature have a huge impact on production [2]. The traditional way of agriculture is not meeting current demands. The reason is a lack of knowledge that acts as a barrier to extracting the information to enhance production. To overcome these problems, smart technology is required to provide strength to farmers to manage their fields cost-effectively and efficiently [3]. The advantages of deep learning (DL) networks in the agriculture sector are shown in Fig. 1. In smart or

intelligent farming, traditional methods combine with technology like sensors, radar, IoT, and machine learning (ML). These technologies enable lowering human costs and improving productivity which can monitor soil, crops, and disease detection. A lot of data gets generated when dealing with intelligent farming, for which ML plays a central role in analyzing the data and making predictions [4].

The data is generated through sensors and other instruments in the form of images, numerical, video, or text form. For analyzing the data there are some popular algorithms that support vector machine, perceptron, clustering, k-means, linear and nonlinear regression, and convolution neural network (CNN) [5]. The agro-industries have been using CNN or other neural networks (NN) to enhance and improve products [2][5] like leaf counting, yield prediction, leaf segmentation[6], predicting leaf stress [7], and crop prediction [8].

Quantum neural network (QNN) has an extended version of quantum convolution network (QCNN) that is the quantum form of NN and CNN. They use quantum superposition and engagement to process the data. This concept is in the developing stage. The main advantage of quantum over classical is that it can perform complex tasks faster in special cases than classical. Initially, QNN and QCNN were developed for quantum information processors and quantum phase recognition[9]. However, there were some limitations to applying DNN to agriculture. Limited data availability, domain adaptation due to highly variable environments, robustness, adaptation of technology by farmers, and transparency states few of the deficits [10].

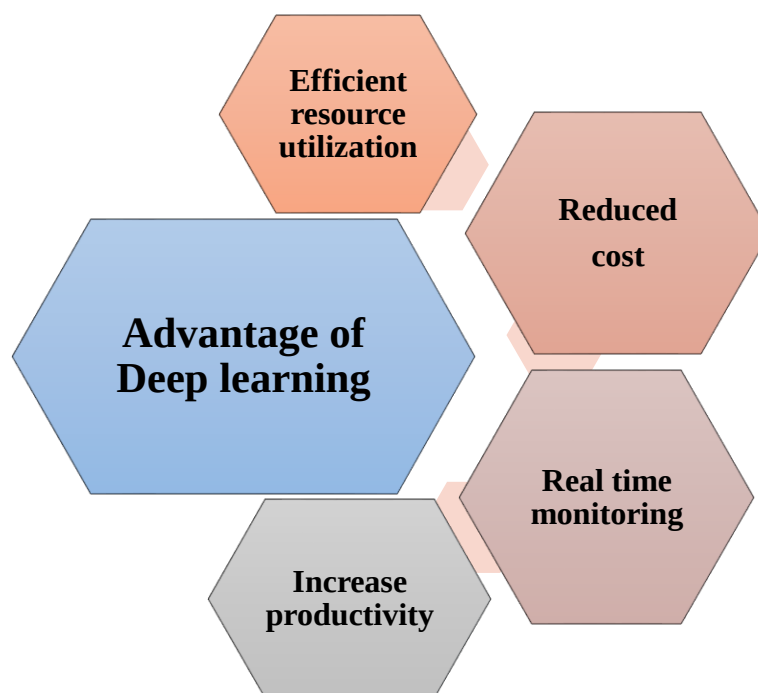


Fig 1. The benefit of deep learning in agriculture

In this paper, we explore the opportunity of classical and quantum DL for the agriculture sector. We use DNN and CNN to predict wheat plant diseases. The accuracy study investigates the performance, efficiency, and applicability of both DL approaches in dealing with plant disease-related challenges and optimizing agricultural output. By evaluating the strengths and limitations of classical and quantum DL techniques, this research seeks to provide valuable

insights for policymakers, researchers, and stakeholders to make informed choices regarding data-driven agricultural strategies.

II. Classical Deep Learning (NN)

a. Introduction: In the digital era, data has become an important asset for actual prediction in many areas. Data can be structured, semi-structured, or unstructured. ML is the art to extract information from data. DL also known as NN is a special case of ML in which the learning algorithm is similar to the modelling of human neurons. NN is very useful in image classification, signal processing, object detection, and pattern identification. CNN is used to extract the selective features from the data [11].

In DL architecture there are two parts one is linear and another is nonlinear which is called the activation function as shown in Fig. 2. Apart from the input and output layers, there are hidden layers. DL can be categorized into feed-forward and back-propagation networks.

$$z_i = \sum_{i=1}^{i=N} (w_i \times x_i) + b_i \quad a_i = \sigma(z_i), a_i = \text{activation function}$$

w_i = weight parameter,

σ = tanh, sigmoid or Relu function

x_i = input parameter, b_i = bias

NN and CNN are optimized by the backpropagation method. The loss function is differentiated concerning weight and bias and its value is updated according to its gradient [12].

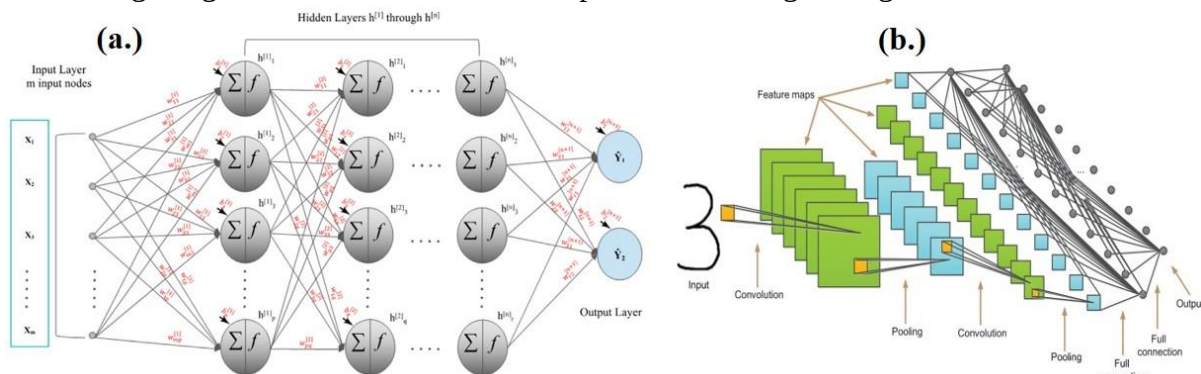


Fig 2. The architecture of neural network. (a.) Deep neural network [13] (b.) Convolution neural network [14].

b. Deep learning in Agriculture: With the growth of the Internet of Things (IoT), nanotechnology-based sensors, drone technology, and high and small computational processor, this sector is growing rapidly. There have been developments like that of the three-stage automatic detection of lesions in coffee plants images captured by smartphones[14][15]. Similarly, a Res-Net v2 model was developed to detect pests and diseases [16]. We have stated a few DL techniques, datasets, and performance in tabular form in Table 1.

Table 1. Deep learning techniques and results in Agriculture application

Field	Dataset library	Method	Accuracy	Reference
Crop yield prediction	Data acquired by Government	Random forest, Support vector machine (SVM), Long term-short term memory (LSTM)	95-98 %	[2]
Soil quality	Kerala state dataset	Extreme learning algorithm	88.59%	[17]
Disease detection on apple leaf	Apple leaf from plant village dataset	SVM, random forest, ML, decision tree, K-nearest neighbour	77.2%	[18]
Plant leaf diseases	Self-acquired by camera	Random forest, SVM, Iterative dichotomiser 3	85-96%	[19]
Plant detection	Plant village dataset	GoogLeNet	99.34%	[20]
Tea Leaf	Self-acquired through mobile camera	Deep CNN	97.04%	[21]
Rice grain	Data collected through camera	Logistic regression, Decision tree, Naïve Bayes, SVM	93-97%	[22]

III. Quantum Machine Learning

Quantum Machine Learning (QML) is a method to process the data on a quantum computer (QC). Quantum computer works on the theory of quantum mechanics. The unit cell of QC is qubits, denoted by $\psi = \alpha|0\rangle + \beta|1\rangle$ where α and $\beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$. In logical operation, quantum is different from classical in terms of reversibility [23]. Unitary matrix a type of square matrix is responsible for reversibility in quantum circuit. Some quantum gates are,

$$\text{Hadamard gate } H = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$$

$$\text{CNOT gate} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix}$$

$$\text{Pauli matrices } \sigma_x = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}, \sigma_y = \begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix}, \sigma_z = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix}$$

QNN is a novel ML model, joining the ranks of Kernel methods, Boltzmann machines, tensor networks, and variation methods. Within QNN, several crucial parameters require consideration. Firstly, the variation circuit entails organizing unitary gates in a specific manner.

This involves applying quantum gates subsequent to the preparation of the initial state, and ultimately, the output is measured. Secondly, quantum circuit learning can be accomplished through simulation-based or hardware-based methods, which utilize gradient descent to optimize the circuit through parameter shift. Thirdly, quantum nodes act as a classical-quantum interface, handling distinct types of information. Fourth, hybrid computation is employed where some aspects of the simulation remain classical, such as optimization tasks and output processing [24] [25].

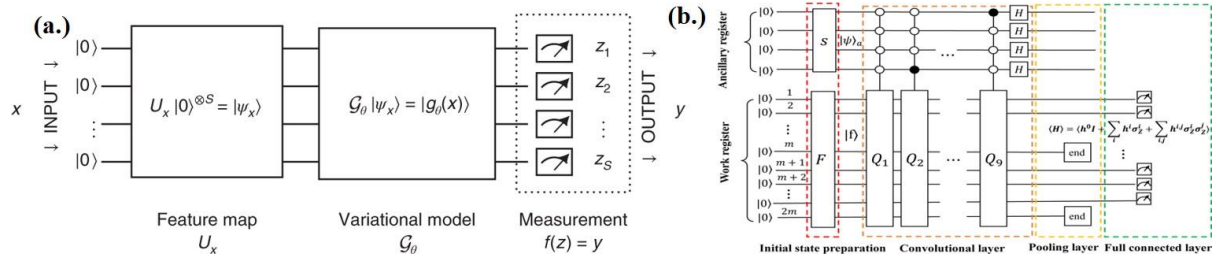


Fig 3. The systematic diagram of quantum circuit, showcasing two distinct models:

(a.) Quantum neural network [24] (b.) Quantum convolution network [25].

As illustrated in Fig. 3 in both models, image data is encoded into a quantum state through a shared feature map. In (a), the data encoding step is followed by a circuit of unitary gates, and subsequently, measurements of the states are performed. In (b), the circuit involves convolution variation followed by pooling layers circuits, and finally, measurements are conducted.

Classical data encoding into quantum states is the foremost and critical step in both QNN and QCNN. Various techniques exist to encode image pixel data into quantum states. In QCNN, the concept is analogous to performing convolution on quantum circuits, and in QML, the data is encoded into quantum states, with QCNN designed specifically to recognize these quantum states [26]. The preparation of the initial quantum state is of utmost importance, as higher entanglement leads to the updating of more independent weights. This highlights the significance of properly initializing the quantum state to achieve desirable outcomes in the training process of QNN and QCNN.

IV. Methodology

The proposed ML models were evaluated on the Kaggle [27] wheat dataset, which consists of 102 healthy and 305 disease sample images. A representative image of healthy and diseased wheat leaf is shown in Fig. 4. Due to the dataset's class imbalance, we first balanced it by employing augmentation techniques such as angular rotation and horizontal and vertical shift. As a result, the final dataset contained 810 images, each with a size of $28 \times 28 \times 3$. To prepare the images for training, we normalized them by dividing each pixel value by 255. For this simulation, we utilize Python 3.7 [28] as the programming language along with the following libraries: tensorflow-keras [29] for traditional ML models, qiskit [30] for QNN, and tensorflow-quantum [31] for QML. These libraries provide the necessary tools and frameworks to build and train the models efficiently. By using this setup, we aimed to address the challenges posed by class imbalance and explore the potential of QNNs in comparison to traditional ML models for image classification tasks on agricultural data.



Fig 4. Images of Wheat leaf used in data processing (a.) Healthy (b.) Septoria diseased (c.) Stripe rust disease

For classical NN, we changed the image size to 28×28 further with three hidden layers and the activation function as Relu. The activation function for the output layer is the sigmoid function. The node in the hidden layer was 784, 500, and 250 with a learning rate of 0.001 and a dropout of 0.35. In NN, the EfficientNet model was considered which was first introduced at ICML 2019 conference [32].

Table 1. Parameter for convolution neural network

Model	EfficientNet
Pooling	Max pooling
Weighted	Image net
Momentum	0.90
Learning rate	0.001
Activation (output)	Softmax
Activation (Dense layer)	Relu
Loss function	Cross-entropy

For QNN implementation, the original 28×28 image size is too large to be directly processed, so a reduction to 4×4 size was performed to fit into the quantum circuit. To convert the classical image data into quantum states, an angular encoding technique was employed. The feature mapping involved applying rotation gate Rx and Hadamard gate to prepare the quantum circuits. The QNN and QCNN circuits were constructed following the references [24], [25], and [26], ensuring the proper design and configuration of the quantum model. By leveraging these quantum circuits, we explored the capabilities of QNN and QCNN for image classification tasks. The implementation process involved adapting the image data into a suitable format for quantum computation, allowing the quantum model to process and extract features effectively.

V. Result

In our investigation of classical and quantum neural networks, we compared the accuracies of four distinct models: CNN, NN, QNN, and QCNN plotted in Fig.5. Surprisingly, the classical CNN model exhibited the highest accuracy, achieving an impressive 91.32% on the test dataset. This outcome is particularly remarkable considering the common belief in quantum advantage for ML tasks. The second-best performing model was the NN with an accuracy of 86.12%. While it did not outperform the classical CNN, its results demonstrated the continued efficacy of classical neural networks in ML applications. QNN, despite its quantum-inspired nature, displayed a lower accuracy of 72.81%. These results indicate that, in the absence of an actual QC platform, the potential quantum advantage of QNN was not fully realized on classical hardware. Similarly, the QCNN achieved an accuracy of 78.53%, falling short of the classical

CNN's performance. This suggests that even though quantum-inspired techniques can be promising, their implementation on classical systems may not always yield the anticipated gains.

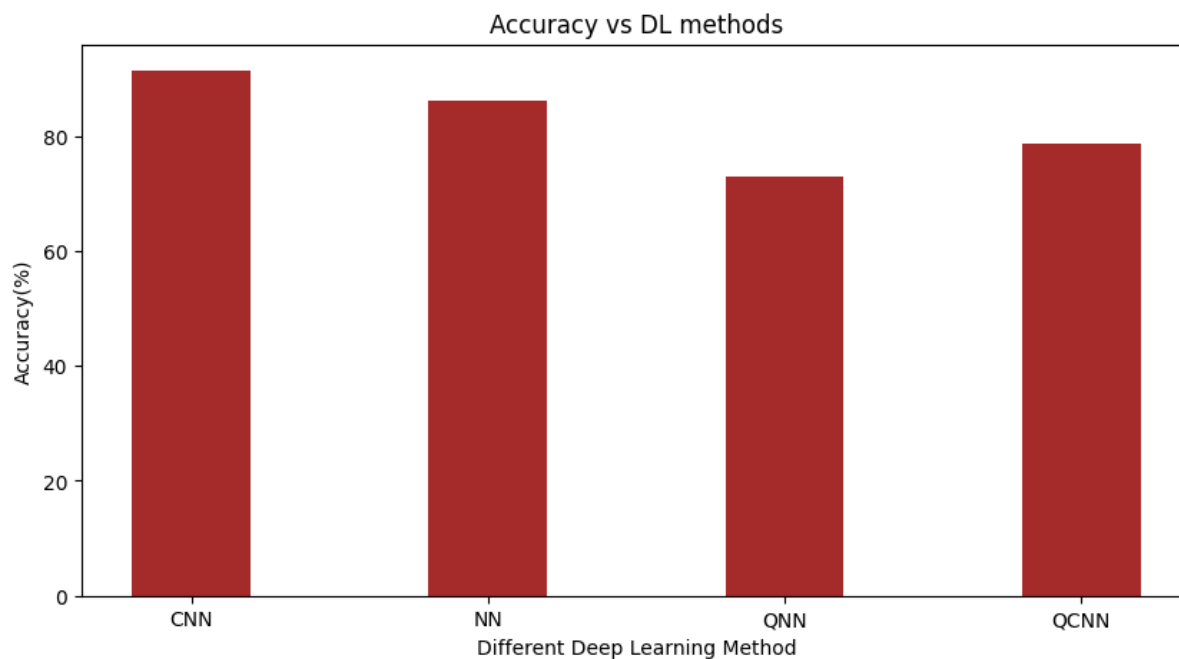


Fig 5. Results of machine learning techniques from classical to quantum methods with accuracy of 'CNN':91.32, 'NN':86.12, 'QNN':72.81, 'QCNN':78.53.

VI. Conclusion

In conclusion, ML continues to evolve and has a profound impact on technology. Although classical ML and IoT have found applications in agro products, the potential of quantum neural networks haven't been much being explored. However, current results indicate that classical ML outperforms quantum ML in accuracy, with CNN achieving 91 % and QNN reaching 72 % accuracy. The hardware used for computations, including GPU support, plays a vital role in influencing the outcomes. The study employed qiskit and TensorFlow-Quantum for QML simulations, with plans to investigate spike neural networks and utilize actual quantum computing hardware in future research. The primary objective of this work is to establish reliable ML models using QNNs to enhance productivity and decision-making performance in the future. While quantum computing in agriculture is still developing, the integration of high-performance systems holds promise for potential advancements in the field.

References:

- [1] V. Meshram, K. Patil, V. Meshram, D. Hanchate, and S. D. Ramkteke, "Machine learning in agriculture domain: A state-of-art survey," *Artif. Intell. Life Sci.*, vol. 1, p. 100010, Dec. 2021, doi: 10.1016/J.AILSCI.2021.100010.
- [2] K. Jhajharia, P. Mathur, S. Jain, and S. Nijhawan, "Crop Yield Prediction using Machine Learning and Deep Learning Techniques," *Procedia Comput. Sci.*, vol. 218, pp. 406–417, Jan. 2023, doi: 10.1016/J.PROCS.2023.01.023.
- [3] S. Goel, K. Guleria, S. N. Panda, S. Goel, K. Guleria, and S. N. Panda, "Machine

- Learning Techniques for Precision Agriculture Using Wireless Sensor Networks,*” ECSTr, vol. 107, no. 1, pp. 9229–9238, Apr. 2022, doi: 10.1149/10701.9229ECST.
- [4] S. Condran, M. Bewong, M. Z. Islam, L. Maphosa, and L. Zheng, “Machine Learning in Precision Agriculture: A Survey on Trends, Applications and Evaluations over Two Decades,” *IEEE Access*, vol. 10, pp. 73786–73803, 2022, doi: 10.1109/ACCESS.2022.3188649.
- [5] K. Dokic, L. Blaskovic, and D. Mandusic, “From machine learning to deep learning in agriculture – the quantitative review of trends,” *IOP Conf. Ser. Earth Environ. Sci.*, vol. 614, no. 1, p. 012138, Dec. 2020, doi: 10.1088/1755-1315/614/1/012138.
- [6] A. Karami, M. Crawford, and E. J. Delp, “Automatic Plant Counting and Location Based on a Few-Shot Learning Technique,” *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 13, pp. 5872–5886, Jan. 2020, doi: 10.1109/JSTARS.2020.3025790.
- [7] V. S. Dhaka et al., “A Survey of Deep Convolutional Neural Networks Applied for Prediction of Plant Leaf Diseases,” *Sensors* 2021, Vol. 21, Page 4749, vol. 21, no. 14, p. 4749, Jul. 2021, doi: 10.3390/S21144749.
- [8] Z. Xiao, X. Xu, H. Xing, F. Song, X. Wang, and B. Zhao, “A federated learning system with enhanced feature extraction for human activity recognition,” *Knowledge-Based Syst.*, vol. 229, Oct. 2021, doi: 10.1016/J.KNOSYS.2021.107338.
- [9] I. Cong, S. Choi, and M. D. Lukin, “Quantum convolutional neural networks,” *Nat. Phys.* 2019 1512, vol. 15, no. 12, pp. 1273–1278, Aug. 2019, doi: 10.1038/s41567-019-0648-8.
- [10] I. Attri, L. K. Awasthi, T. P. Sharma, and P. Rathee, “A review of deep learning techniques used in agriculture,” *Ecol. Inform.*, vol. 77, p. 102217, Nov. 2023, doi: 10.1016/J.ECOINF.2023.102217.
- [11] L. Alzubaidi et al., “Review of deep learning: concepts, CNN architectures, challenges, applications, future directions,” *J. Big Data* 2021 81, vol. 8, no. 1, pp. 1–74, Mar. 2021, doi: 10.1186/S40537-021-00444-8.
- [12] Aggarwal, Charu C. *Neural Networks and Deep Learning: A Textbook*. Springer Cham, 2018. <https://doi.org/10.1007/978-3-319-94463-0>
- [13] X. Jiang and C. Xu, “Deep Learning and Machine Learning with Grid Search to Predict Later Occurrence of Breast Cancer Metastasis Using Clinical Data,” *J. Clin. Med.*, vol. 11, no. 19, p. 5772, Oct. 2022, doi: 10.3390/JCM11195772/S1.
- [14] T. Saranya, C. Deisy, S. Sridevi, and K. S. M. Anbananthen, “A comparative study of deep learning and Internet of Things for precision agriculture,” *Eng. Appl. Artif. Intell.*, vol. 122, p. 106034, Jun. 2023, doi: 10.1016/J.ENGAPPAI.2023.106034.
- [15] C. J. Chen, Y. Y. Huang, Y. S. Li, C. Y. Chang, and Y. M. Huang, “An AIoT Based Smart Agricultural System for Pests Detection,” *IEEE Access*, vol. 8, pp. 180750–180761, 2020, doi: 10.1109/ACCESS.2020.3024891.
- [16] Y. Ai, C. Sun, J. Tie, and X. Cai, “Research on recognition model of crop diseases and insect pests based on deep learning in harsh environments,” *IEEE Access*, vol. 8, pp. 171686–171693, 2020, doi: 10.1109/ACCESS.2020.3025325.
- [17] M. S. Suchithra and M. L. Pai, “Improving the prediction accuracy of soil nutrient classification by optimizing extreme learning machine parameters,” *Inf. Process. Agric.*, vol. 7, no. 1, pp. 72–82, Mar. 2020, doi: 10.1016/J.INPA.2019.05.003.

- [18] M. Francis, K. S. M. Anbananthen, D. Chelliah, S. Kannan, S. Subbiah, and J. Krishnan, "Smart Farm-Care using a Deep Learning Model on Mobile Phones," *Emerg. Sci. J.*, vol. 7, no. 2, pp. 480–497, Feb. 2023, doi: 10.28991/ESJ-2023-07-02-013.
- [19] A. S. Zamani et al., "Performance of Machine Learning and Image Processing in Plant Leaf Disease Detection," *J. Food Qual.*, vol. 2022, 2022, doi: 10.1155/2022/1598796.
- [20] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Front. Plant Sci.*, vol. 7, no. September, p. 215232, Sep. 2016, doi: 10.3389/FPLS.2016.01419/BIBTEX.
- [21] S. Datta and N. Gupta, "A Novel Approach For the Detection of Tea Leaf Disease Using Deep Neural Network," *Procedia Comput. Sci.*, vol. 218, pp. 2273–2286, Jan. 2023, doi: 10.1016/J.PROCS.2023.01.203.
- [22] B. Arora, N. Bhagat, L. Saritha, and S. Arcot, "Rice Grain Classification using Image Processing & Machine Learning Techniques," *2020 Int. Conf. Inven. Comput. Technol.*, pp. 205–208, Feb. 2020, doi: 10.1109/ICICT48043.2020.9112418.
- [23] M. Schuld, I. Sinayskiy, and F. Petruccione, "An introduction to quantum machine learning," <http://dx.doi.org/10.1080/00107514.2014.964942>, vol. 56, no. 2, pp. 172–185, Apr. 2014, doi: 10.1080/00107514.2014.964942.
- [24] A. Abbas, D. Sutter, C. Zoufal, A. Lucchi, A. Figalli, and S. Woerner, "The power of quantum neural networks," *Nat. Comput. Sci.*, vol. 1, no. 6, pp. 403–409, Jun. 2021, doi: 10.1038/S43588-021-00084-1.
- [25] K. Beer et al., "Training deep quantum neural networks," *Nat. Commun.* 2020 111, vol. 11, no. 1, pp. 1–6, Feb. 2020, doi: 10.1038/s41467-020-14454-2.
- [26] M. Henderson, S. Shakya, S. Pradhan, and T. Cook, "Quantum evolutionary neural networks: powering image recognition with quantum circuits," *Quantum Mach. Intell.*, vol. 2, no. 1, pp. 1–9, Jun. 2020, doi: 10.1007/S42484-020-00012-Y/METRICS.
- [27] Kaggle. [Online] Holeta wheat farm, Ethiopia. <https://www.kaggle.com/datasets/olyadgetch/wheat-leaf-dataset>.
- [28] Python. [Online] Python Software Foundation. <https://www.python.org/>.
- [29] Tensor flow. [Online] Google. <https://www.tensorflow.org/guide/keras>.
- [30] Qiskit. [Online] IBM. <https://qiskit.org/>.
- [31] Tensor flow-quantum. [Online] Google. <https://www.tensorflow.org/quantum>.
- [32] EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. Mingxing Tan, Quoc V. Le. 2019. ICML 2019.